EXPERIMENT 13

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Aim

To prepare an algorithm and test-case table, then write, compile, execute, and test a menu-driven C program using **switch-case** to perform operations on the even and odd digits of a positive integer.

Problem Statement

Input a positive integer and choose: **(1)** count the number of even and odd digits, or **(2)** find the sum of the even digits and the sum of the odd digits separately.

Algorithm

1. Start and input a positive integer n.
2. If n <= 0, display an error message and stop.
3. Display the menu and input the user's choice.
4. Use switch(choice).
Case 1: Set even_count = 0 and odd_count = 0. Copy n to temp. Extract each digit using temp % 10. If digit % 2 == 0, increment even_count; otherwise increment odd_count. Divide temp by 10 and repeat until all digits are processed. Display both counts.
Case 2: Set sum_even = 0 and sum_odd = 0. Copy n to temp. Extract each digit. If it is even, add it to sum_even; otherwise add it to sum_odd. Continue until all digits are processed. Display both sums.
Default: Display Invalid choice.
5. Stop.

Test Case Table

Test	Number	Choice	Expected Result
1	12045	1	Even digits = 3, Odd digits = 2
2	12345	1	Even digits = 2, Odd digits = 3
3	24680	1	Even digits = 5, Odd digits = 0
4	12345	2	Sum_even = 6, Sum_odd = 9
5	24681	2	Sum_even = 20, Sum_odd = 1
6	13579	2	Sum_even = 0, Sum_odd = 25
7	123	3	Invalid choice

C Program

```
#include <stdio.h>

int main()
{
    int n, temp, choice, digit;
    int even_count = 0, odd_count = 0;
    int sum_even = 0, sum_odd = 0;

    printf("Enter a positive integer: ");
    scanf("%d", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            temp = n;

            while (temp > 0) {
                digit = temp % 10;

                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;

                temp = temp / 10;
            }

            printf("Even digits = %d\n", even_count);
            printf("Odd digits = %d\n", odd_count);
            break;

        case 2:
            temp = n;

            while (temp > 0) {
                digit = temp % 10;

                if (digit % 2 == 0)
                    sum_even = sum_even + digit;
                else
                    sum_odd = sum_odd + digit;

                temp = temp / 10;
            }

            printf("Sum_even = %d\n", sum_even);
            printf("Sum_odd = %d\n", sum_odd);
            break;

        default:
            printf("Invalid choice.\n");
    }

    return 0;
}
```

Compilation and Execution (Linux/GCC)

Save as: `experiment13.c`

Compile: `gcc experiment13.c -o experiment13`

Execute: `./experiment13`

Sample Executions and Testing

Test 1 - Choice 1

```
Enter a positive integer: 12045
Enter your choice: 1
Even digits = 3
Odd digits = 2
```

Result: PASS

Test 2 - Choice 2

```
Enter a positive integer: 12345
Enter your choice: 2
Sum_even = 6
Sum_odd = 9
```

Result: PASS

Result

The menu-driven C program was successfully prepared using switch-case, compiled, executed, and tested against the prepared test cases. It correctly counts even/odd digits and calculates their separate sums.